

Lukas Brunner

curriculum vitae

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Current Position

since **Senior researcher**
05/2024 Cluster of Excellence Climate, Climatic Change, and Society (CLICCS) and Climate Extremes group, University of Hamburg

Previous Positions

01/2022 - **Senior scientist**
04/2024 Group for Climate Dynamics and Modeling, Department of Meteorology and Geophysics, University of Vienna
05/2023 - **Visiting scientist**
06/2023 Institute of Meteorology, FU Berlin
10/2020 - **Senior scientist**
12/2021 Climate Physics group, Institute for Atmospheric and Climate Science, ETH Zurich
05/2018 - **Postdoctoral scientist**
09/2020 Climate Physics group, Institute for Atmospheric and Climate Science, ETH Zurich
04/2017 - **Visiting scientist**
09/2017 Center for International Climate Research Oslo (CICERO)
10/2015 - **Visiting scientist**
03/2016 Chair of Climate System Science, School of Geosciences, University of Edinburgh
10/2014 - **Predoctoral scientist**
03/2018 Research group ARSCliSys, Wegener Center, University of Graz

Education

06/2021 **Certificate of Open Studies: Applied Data Science Machine Learning**
EPFL Extension School, Switzerland
03/2018 **Dr. rer. nat. in Environmental System Sciences**, University of Graz, Austria
Thesis A new perspective on atmospheric blocking from observations – detection, analysis, and impacts
07/2014 **MSc in Geophysics**, University of Graz, Austria
Thesis Stratospheric ozone and temperature evolution over the past decades
10/2012 **BSc in Physics**, University of Graz, Austria
06/2008 School leaving examination

Parental Leave

I was on parental leave from April to June 2026. In February and March 2026, I was on partial parental leave and worked 50 %.

Funding, Scholarships, and Awards

- 2024 **High-resolution machine learning for the climate community in Austria**
- 2023 **Wissenschaftspreis des Landes Vorarlberg for young researchers**
- 2022 **Scholarship as youth representative of the European Forum Alpbach at COP27**
- 2018 **Performance scholarship for my PhD project**
- 2017-2018 **Member of PRO SCIENTIA – Österreichisches Studienförderungswerk**
- 2016-2018 **Scholarships for the European Forum Alpbach (x3)**
- 2016 **Marietta Blau scholarship from the Austrian Exchange Service**

Committee Work, Convening, and Volunteering

- since 2026 **Member of the WMO/WCRP climate indices review group**
- since 2026 **Member of the CLICCS Diversity, Equity, and Inclusion (DEI) Board**
- since 2024 **Member of the WCRP activity Atmospheric Temperature Changes and their Drivers**
- 2025 **Team lead at the WCRP global km-scale hackathon**
WCRP Digital Earths Lighthouse Activity
- 2025 **Session convener at EGU25**
 - New developments in Earth system modelling: model evaluation, benchmarking, and constraining
 - Up-Goer Five Challenge: Making Big Ideas Simpler by Talking About Them in Words We Use a Lot
- since 2024 **Member of the Scientific Advisory Board of the Austrian Climate Scenarios**
- 2024 **Session convener at EGU24**
Regional Climate Modelling, Including CORDEX and Constraining Global Multi-Model Ensembles
- 2024 **Mentor in the EGU mentoring programme**
- 2023-2024 **Member of the University of Vienna Sustainability Advisory Board**
- 2022-2024 **Member of the Steering Committee of the Austrian Climate Scenarios**
- 2021 - **Climate expert and participant at the Future of the Earth performance**
- 2022 **The Future of the Earth interactive theatre performance**
- 2022 **Session chair CliMatters workshop**
- 2019-2021 **Board member (president in 2021) of Club Alpbach Vorarlberg**
My work included raising funding for up to 15 scholarships for the European Forum Alpbach together with a small team, organising events at the forum and in the region of Vorarlberg during the year, and leading an alumni club with more than 100 members
- since 2018 **Reviewer for scientific journals**
More than 50 verified reviews on Web of Science

Selected Presentations

- 07/2026 **km-scale global modelling summit, Hamburg**
- 04/2025 **WCRP APARC ATC spring meeting, Graz**
- 04/2025 **MPI-M/University of Hamburg, Klimacampus Kolloquium, Hamburg (invited)**

03/2025 University of Bern, Colloquium in Atmosphere, Climate and Earth Observation, Bern (invited)

01/2025 University of Innsbruck, ACINN Graduate Seminar, Innsbruck (invited)

10/2024 NCKF Climate Research Symposium, Copenhagen (keynote)

07/2024 University of Freiburg, Freiburg (invited)

03/2024 Deutsche Klimatagung, Potsdam (keynote)

02/2024 Climate Analytics, Berlin (invited)

07/2023 IUGG General Assembly, Berlin

05/2023 Institute for Meteorology colloquium, Leipzig University (invited)

05/2023 Institute for Meteorology colloquium, FU Berlin

05/2023 Climate Extremes group, University of Hamburg (invited)

03/2023 Global Carbon Budget workshop (keynote)

01/2022 Department of Meteorology and Geophysics colloquium, University of Vienna

11/2021 EC-Earth Consortium General Assembly 2021 (keynote)

10/2021 Wegener Center Common Space, University of Graz, Graz (invited)

04/2017 EGU General Assembly, Vienna

12/2015 AGU Fall Meeting, San Francisco

Supervision and Teaching

Teaching

since 2024 **Weather and Climate Extremes in a Changing Climate**
Master course, 2 semester hours (SH), University of Hamburg (part of the course). I developed several lectures and interactive exercises for this course.

2025 **Grundlagen des Klimawandels**
Bachelor course, 2 SH, University of Hamburg (part of the course)

2024 **Climate Modelling**
Master course, 4 SH, University of Hamburg (part of the course)

2023-2024 **Paper Club**
Master course, 2 SH, University of Vienna. I developed and taught this course together with Blaž Gasparini.

2022-2024 **Numerische Methoden**
Bachelor course, 2 SH, University of Vienna. I developed and taught this course.

2022-2024 **Modelling and Data Analysis**
Master course, 2 SH, University of Vienna. I developed and taught this course together with Marina Dütsch. We also developed the accompanying exercises, which were taught by a PhD student.

2022-2024 **Exercises in Introduction to Computational Meteorology**
Master course, 2 SH, University of Vienna. I developed and taught this course.

Supervision

2026 **Using km-scale global climate model simulations to study the physical impacts of climate change for informing adaptation strategies**
Supervision of PhD thesis with Jana Sillmann

- 2026 **Effect of an AMOC shutdown on extremes in Europe using km-scale models**
Supervision of Master's thesis with Keno Riechers (MPI-M)
- 2025 **Spatiotemporal Roles of Heatwave Drivers in Europe (completed)**
Supervision of Master's thesis with Jana Sillmann
- 2025 **Separating Sources of Uncertainty in Future Climate Extremes (completed)**
Supervision of Master's thesis with Jana Sillmann
- 2024 **Effect of climate variability and extremes on heating and cooling energy demand in Canada (completed)**
Supervision of Master's thesis with Jana Sillmann
- 2024 **Exploring Classifier Skill in Distinguishing Climate Model and Observational Data (completed)**
Supervision of Master's thesis
- 2024 **Changes in heatwave properties under climate change (completed)**
Supervision of Master's thesis with Aiko Voigt
- 2024 **Extracting forced signals of climate change from daily temperature patterns across scales with CESM2 (completed)**
Supervision of Bachelor's thesis
- 2024 **Analyse eines Fehlers in der Definition der Hitzemetrik Cumulative Heat (completed)**
Supervision of Bachelor's thesis
- 2021 **Interannual to decadal precipitation variability in a warming climate: spatial structure and underlying mechanisms (completed)**
Supervision of Master's thesis with Flavio Lehner
- 2020 **Assessing climate model uncertainties – an encounter with CMIP6 (completed)**
Supervision of Bachelor's thesis
- 2019 **Changes in northern hemisphere blocking occurrence under $2\times\text{CO}_2$ based on CESM (completed)**
Supervision of Bachelor's thesis

Selected Outreach Activities

- 2025-2026 Consultant in the development of the Reuters Climate Monitor
<https://www.reuters.com/graphics/CLIMATE-AUTOMATED/MONITOR/akpeykqqapr/>
- 10/2025 Public lecture: Naturwissenschaftliche Grundlagen der Klimakrise
<https://youtu.be/IuTXbtR1fFw>
- 06/2025 Public lecture: Wie ändern sich Wetterextreme mit dem Klimawandel?
<https://youtu.be/0KGi4Y6J2Ls>
- 2025-2026 School lectures: Wetter- und Klimaextreme im Kontext des Klimawandels (x2)
<https://www.uni-hamburg.de/wissen-fuer-alle/wir-wollens-wissen.html>
- 2024 Mentioned in Media: Die Presse
"Irrtum der Experten: Es gibt mehr Hitzetage", <https://www.diepresse.com/18290185/irrtum-der-experten-es-gibt-mehr-hitzetage>
- 10/2024 Guest lecture: Naturwissenschaftliche Grundlagen der Klimakrise
Klimarecht im intra- und interdisziplinären Dialog, University of Vienna

- 2022 Article: Rudolphina
<https://rudolphina.univie.ac.at/klimakonferenz-cop27-hitze-duerre-flut-extremwetter-katastrophen>
- 2021 Climate expert at the Future of the Earth interactive theatre performance, Basel
- 08/2023 Open debate: Data sparks action – the role of information in tackling the climate crisis
 European Forum Alpbach, Austria
- 2020 Mentioned in media: Ars Technica
<https://arstechnica.com/science/2020/11/newest-climate-models-shouldnt-raise-future-warming-projections>
- 2023-2024 Guest lectures: Naturwissenschaftliche Grundlagen der Klimakrise (x2)
 Green Finance Seminar, University of Vienna
- 07/2022 Participation in the Vienna Kinderuni
- 2017 Mentioned in media: Der Standard (in German)
<https://www.derstandard.at/story/2000051806269/atmosphaerische-blockaden-fuehren-zu-kaelteeinbruechen-im-fruehjahr>
- 10/2022 Public lecture: Naturwissenschaftliche Grundlagen der Klimakrise
 Inspirationstag der Katholischen Kirche Wien
- 07/2022 Interview in Ö1 public radio: Wie entstehen Wolken?

Publications

Upcoming

- accepted **Brunner L.**, Mithal V., Borchert L. F., Sillmann J., and Poschlod B.: The Variability Atlas: How internal climate variability affects the estimation of climate extreme indices, *Environ. Res. Lett.*
- accepted Aglas-Leitner P., Perkins-Kirkpatrick S. E., **Brunner L.**, Undorf S., Aglas-Leitner F., Fischer E., and Stone D. A.: Recent extreme heatwaves are no match for future events, *Nature Communications*
- in review Poschlod B., **Brunner L.**, and Sillmann J.: LESSONS report: Fine-scale representation of consecutive dry days, *EGUsphere*, DOI: <https://doi.org/10.5194/egusphere-2026-4014>
- in review Uckan Y., **Brunner L.**, Sillmann J., and Kang S. M.: Influence of Southern Ocean temperature trend biases on fall cold extremes in North America, *Clim. Dyn.*, DOI: <https://doi.org/10.21203/rs.3.rs-9610006/v1>
- in review van Heerwaarden C. C., Veerman M. A., Benedict I., **Brunner L.**, Dolores-Tesillos E., Dutra E., Fischer E., Lee J., Martius O., Pedruzo-Bagazgoitia X., Proske U., Warnau S. N., Wille J. D., and Hohenegger C.: Evaluating local climate in global storm-resolving models with the Köppen-Geiger classification, *Environ. Res. Lett.*, DOI: <https://doi.org/10.48550/arXiv.2604.25447>
- in review Castillo Bautista N., Gaetani M., Borchert L. F., Poschlod B., **Brunner L.**, Sillmann J., and Martina M. L. V.: Large-scale drivers of compounding hot and dry events in three breadbasket regions, *EGUsphere*, DOI: <https://doi.org/10.5194/egusphere-2025-5073>

Peer-Reviewed

- 08/2026 Qu S., Sillmann J., Barrett B. W., Graffy P. M., Poschlod B., **Brunner L.**, Mansour R., von Szombathely M., Hay-Chapman F., and Horton T. H.: Integrating machine learning-based variable selection into Heat Vulnerability Index design, *Environ. Res. Health*, DOI: <https://doi.org/10.1088/2752-5309/ae9633>
- 04/2026 **Brunner L.**, Ghosh R., Haimberger L., Hohenegger C., Putrasahan D., Rackow T., Knutti R., and Voigt A.: Three decades of simulating global temperature patterns with coupled global climate models, *Commun. Earth Environ.*, DOI: <https://doi.org/10.1038/s43247-026-03497-w>
- 02/2026 Holtanová E., Koláček J., and **Brunner L.**: A global perspective on past and future change in regional seasonal cycle shape, *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-17-181-2026>
- 02/2026 Yoon A., Hohenegger C., Bao J., and **Brunner L.**: Extreme events in the Amazon after deforestation, *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-17-167-2026>
- 10/2025 Segura H. et al. (including **Brunner L.**): nextGEMS: entering the era of kilometer-scale Earth system modeling, *Geosci. Model Dev.*, DOI: <https://doi.org/10.5194/gmd-18-7735-2025>
- 09/2025 Maher N., Phillips A. S., Deser C., Inglin Wills R. C., Lehner F., Fasullo J., Caron J. M., **Brunner L.**, Beyerle U., and Jeffree J.: The updated Multi-Model Large Ensemble Archive and the Climate Variability Diagnostics Package: New tools for the study of climate variability and change, *Geosci. Model Dev.*, DOI: <https://doi.org/10.5194/gmd-18-6341-2025>

- 06/2025 **Brunner L.**, Poschlod B., Dutra E., Fischer E. M., Martius O., and Sillmann J.: A global perspective on the spatial representation of climate extremes from km-scale models, *Environ. Res. Lett.*, DOI: <https://doi.org/10.1088/1748-9326/ade1ef>
- 05/2025 Sobolowski S., Somot S., Fernandez J., Evin G., Brands S., Maraun D., Kotlarski S., Jury M., Benestad R. E., Teichmann C., Christensen O. B., Bülow K., Buonomo E., Katragkou E., Steger C., Sørland S., Nikulin G., McSweeney C., Dobler A., Palmer T., Wilcke R., Boé J., **Brunner L.**, Ribes A., Qasmi S., Nabat P., Sevault F., and Oudar T.: GCM Selection & Ensemble Design: Best Practices and Recommendations from the EURO-CORDEX Community, *Bull. Am. Meteorol. Soc.*, DOI: <https://doi.org/10.1175/BAMS-D-23-0189.1>
- 11/2024 Nguyen V. D., Vorogushyn S., Nissen K., **Brunner L.**, and Merz B.: A non-stationary climate-informed weather generator for assessing future flood risks, *Adv. Stat. Clim. Meteorol. Oceanogr.*, DOI: <https://doi.org/10.5194/ascmo-10-195-2024>
- 04/2024 O'Reilly C., **Brunner L.**, Qasmi S., Nogherotto R., Ballinger A., Booth B., Befort D., Knutti R., Schurer A., Ribes A., Weisheimer A., Coppola E., and McSweeney C.: Assessing observational constraints on future European climate in an out-of-sample framework, *npj Clim. Atmos. Sci.*, DOI: <https://doi.org/10.1038/s41612-024-00648-8>
- 03/2024 **Brunner L.** and Voigt A.: Pitfalls in diagnosing temperature extremes, *Nature Communications*, DOI: <https://doi.org/10.1038/s41467-024-46349-x>
- 08/2023 Merrifield A. L., **Brunner L.**, Lorenz R., Humphrey V., and Knutti R.: Climate model Selection by Independence, Performance, and Spread (ClimSIPS v1.0.1) for regional applications, *Geosci. Model Dev.*, DOI: <https://doi.org/10.5194/gmd-16-4715-2023>
- 07/2023 **Brunner L.** and Sippel S.: Identifying climate models based on their daily output using machine learning, *Environ. Data Sci.*, DOI: <https://doi.org/10.1017/eds.2023.23>
- 04/2023 Palmer T. E., McSweeney C. F., Booth B. B. B., Priestley M. D. K., Davini P., **Brunner L.**, Borchert L., and Menary M. B.: Performance based sub-selection of CMIP6 models for impact assessments in Europe, *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-14-457-2023>
- 10/2022 Gründemann G. J., van de Giesen N., **Brunner L.**, and van der Ent R.: Rarest rainfall events will see the greatest relative increase in magnitude under future climate change, *Commun. Earth Environ.*, DOI: <https://doi.org/10.1038/s43247-022-00558-8>
- 06/2022 Befort D. J., **Brunner L.**, Borchert L. F., O'Reilly C. H., Mignot J., Ballinger A. P., Hegerl G. C., and Murphy J. M.: Combination of decadal predictions and climate projections in time: Challenges and potential solutions, *Geophys. Res. Lett.*, DOI: <https://doi.org/10.1029/2022GL098568>
- 11/2021 Sperna Weiland F. C., Visser R. D., Greve P., Bisselink B., **Brunner L.**, and Weerts A. H.: Estimating Regionalized Hydrological Impacts of Climate Change Over Europe by Performance-Based Weighting of CORDEX Projections, *Front. Water*, DOI: <https://doi.org/10.3389/frwa.2021.713537>
- 06/2021 Hegerl G., Ballinger A. P., Booth B., Borchert L. F., **Brunner L.**, Donat M., Doblas-Reyes F., Harris G., Lowe J., Mahmood R., Mignot J., Murphy J., Swingedouw D., and Weisheimer A.: Toward Consistent Observational Constraints in Climate Predictions and Projections, *Front. Clim.*, DOI: <https://doi.org/10.3389/fclim.2021.678109>

- 11/2020 **Brunner L.**, Pendergrass A. G., Lehner F., Merrifield A. L., Lorenz R., and Knutti R.: Reduced global warming from CMIP6 projections when weighting models by performance and independence, *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-11-995-2020>
- 09/2020 **Brunner L.**, McSweeney C., Ballinger A. P., Befort D. J., Benassi M., Booth B. B., Coppola E., de Vries H., Harris G., Hegerl G. C., Knutti R., Lenderink G., Lowe J., Nogherotto R., O'Reilly C., Qasmi S., Ribes A., Stocchi P., and Undorf S.: Comparing Methods to Constrain Future European Climate Projections Using a Consistent Framework, *J. Climate*, DOI: <https://doi.org/10.1175/jcli-d-19-0953.1>
- 09/2020 Merrifield A. L., **Brunner L.**, Lorenz R., and Knutti R.: An investigation of weighting schemes suitable for incorporating large ensembles into multi-model ensembles, *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-11-807-2020>
- 05/2020 Lehner F., Deser C., Maher N., Marotzke J., Fischer E., **Brunner L.**, Knutti R., and Hawkins E.: Partitioning climate projection uncertainty with multiple Large Ensembles and CMIP5/6, *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-11-491-2020>
- 09/2019 **Brunner L.**, Lorenz R., Zumwald M., and Knutti R.: Quantifying uncertainty in European climate projections using combined performance-independence weighting, *Environ. Res. Lett.*, DOI: <https://doi.org/10.1088/1748-9326/ab492f>
- 06/2018 Unterberger C., **Brunner L.**, Nabernegg S., Steininger K., Steiner A. K., Stabentheiner E., Monschein S., and Truhetz H.: Spring frost risk for regional apple production under a warmer climate, *PLoS ONE*, DOI: <https://doi.org/10.1371/journal.pone.0200201>
- 01/2018 **Brunner L.**, Schaller N., Anstey J., Sillmann J., and Steiner A. K.: Dependence of present and future European temperature extremes on the location of atmospheric blocking, *Geophys. Res. Lett.*, DOI: <https://doi.org/10.1029/2018GL077837>
- 01/2017 **Brunner L.** and Steiner A. K.: A global perspective on atmospheric blocking using GPS radio occultation – one decade of observations, *Atmos. Meas. Tech.*, DOI: <https://doi.org/10.5194/amt-10-4727-2017>
- 11/2016 **Brunner L.**, Hegerl G. C., and Steiner A. K.: Connecting atmospheric blocking to European temperature extremes in spring, *J. Climate*, DOI: <https://doi.org/10.1175/JCLI-D-16-0518.1>
- 04/2016 **Brunner L.**, Steiner A. K., Scherllin-Pirscher B., and Jury M. W.: Exploring atmospheric blocking with GPS radio occultation observations, *Atmos. Chem. Phys.*, DOI: <https://doi.org/10.5194/acp-16-4593-2016>

Other Publications

- 03/2020 **Brunner L.**, Hauser M., Lorenz R., and Beyerle U.: The ETH Zurich CMIP6 next generation archive: technical documentation, DOI: <http://doi.org/10.5281/zenodo.3734128>
- 06/2018 Mohankumar S. E. P., Mintz-Woo K., Damert M., **Brunner L.**, and Eise J.: Blogging Climate Change: A Case Study, In: Addressing the Challenges in Communicating Climate Change Across Various Audiences, DOI: <https://doi.org/10.1007/978-3-319-98294-6>
- 04/2018 **Brunner L.**: A new perspective on atmospheric blocking from observations – detection, analysis, and impacts (Dissertation). Wegener Center Verlag Graz, Scientific Report Nr. 76-2018, URL: <https://wegcwww.uni-graz.at/publ/wegcreports/2018/WCV-SciRep-No76-LBrunner-Jun2018.pdf>

06/2014 **Brunner L.:** Stratospheric ozone and temperature evolution over the past decades (Master's thesis). Wegener Center Verlag Graz, Scientific Report Nr. 59-2014, URL: http://wegcwww.uni-graz.at/publ/wegcreports/2014/WCV-SciRep-No59-LBrunner-Aug2014_1.pdf

Published Code and Data

- Tool **Brunner L.** (2026): Variability Atlas for ETCCDI climate extreme indices, https://lukasbrunner.github.io/variability_atlas/
- Dataset & Code **Brunner L.** (2026): Data and code for: Three decades of simulating global temperature patterns with coupled global climate models, <https://doi.org/10.5281/zenodo.18853482>
- Dataset **Brunner L.** (2025): Annual ETCCDI Extreme Indices For ICON-Sapphire And IFS-FESOM (nextGEMS Cycle 4), https://doi.org/10.26050/WDCC/nextGEMS_ExtrInd
- Code **Brunner L.** (2025): Code for: A global perspective on the spatial representation of climate extremes from km-scale models
<https://doi.org/10.5281/zenodo.15619824>
https://github.com/lukasbrunner/Brunner_etal_2025_ERL
- Code **Brunner L.** (2024): Code for: Pitfalls in diagnosing temperature extremes, https://github.com/lukasbrunner/running_window_bias
- Dataset & Code **Brunner L.** and Sippel S. (2023): Data and code for: Identifying climate models based on their daily output using machine learning, <https://doi.org/10.5281/zenodo.7998436>
- Dataset Liu Y., Kalverla P., Alidoost F., Verhoeven S., Vreede B., Booth B., Coppola E., Nogherotto R., **Brunner L.**, Harris G., Qasmi S., Ballinger A., Hegerl G., McSweeney C., O'Reilly C., Palmer T., Ribes A., and de Vries H. (2021): Pre-processed data of atlas in EUCP-WP2 (1.0.1). <https://doi.org/10.5281/zenodo.5679560>
- Code **Brunner L.** et al. (2020): Climate Model Weighting by Independence and Performance (ClimWIP)
<https://github.com/lukasbrunner/ClimWIP> (collaborative project)
https://docs.esmvaltool.org/en/latest/recipes/recipe_climwip.html
- Code **Brunner L.** (2018): A global blocking detection algorithm
<https://github.com/lukasbrunner/blocking>